Data-driven evaluation of ^{56}Fe elastic-scattering angular distributions and covariances

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Abstract. ^{56}Fe is the main isotope of the steel between core and pressure vessel, and its elastic-scattering angular distribution contributes to fast-neutron transport to the vessel and to the uncertainty of the predicted fluence. The covariances available for it in the evaluated libraries are sparse and differ widely between them: that of JEFF-4.0 is inconsistent with the angular distribution it accompanies, and that of JENDL-5 covers a single Legendre order. In the 847 keV to 4 MeV range studied here, both libraries also anchor their central values on two campaigns, out of the 67 experiments available in EXFOR. We present a purely data-driven evaluation of the Legendre coefficients and their covariances over that range, fitted per energy bin by regularized generalized least squares with each experiment's normalization treated as a shared systematic. Two Monte Carlo passes supply the variances and the correlations across order and energy. The resulting 526-group `MF34` sits closer to the experimental database than either library under an identical metric.

1 Introduction

Fast-neutron fluence at the reactor pressure vessel governs embrittlement assessments for pressurized water reactor life extension, and its prediction depends on the nuclear data of the steel between core and vessel [1]. ^{56}Fe is the principal isotope of that steel. Alongside the hydrogen and oxygen of the water, its elastic scattering shapes how fast neutrons cross the steel regions, and in deep-penetration geometry the angular distribution matters as much as the cross section: forward scattering carries neutrons outward, backward scattering returns them toward the core.

For ^{56}Fe this covariance is sparse and differs widely between libraries. Table 1 summarizes the situation in the 847 keV–4 MeV window, where the elastic cross section fluctuates strongly and optical-model calculations do not reproduce the resonance structure, leaving experiment as the primary source of information. JEFF-4.0 [2] carries no `MF34` of its own for this reaction: the covariance is inherited from JEFF-3.3 [3], itself carried over from JEFF-3.0 (2005), and predates the current `MF4`. ENDF/B-VIII.1 [4], which shares the JEFF-4.0 `MF4`, carries none. JENDL-5 [5] provides covariances only for the first Legendre order, on four broad energy groups, whereas the nominal `MF4` grid holds more than a thousand energies in the same interval.

The distribution of the variance across orders is a second problem. Because each library concentrates its variance in a few orders, the propagated band on the differential cross section takes the angular shape of those orders, not that of the disagreement between measurements. Figure 1 shows one example of each artifact, and both recur across the window. The large high-order variance of JEFF-4.0 gives an oscillatory, lobed band, here from $\ell = 6$ at 2.8 MeV. The single $\ell = 1$ term of JENDL-5 gives, at every energy, a dipolar band that collapses to zero at $\cos\theta = 0$, where the data are no better known than elsewhere.

Both libraries also take their central values from two campaigns, Kinney 1976 [6] below 2.5 MeV and Smith 1980 [7] above it; for JEFF-4.0 this choice comes from the CIELO evaluation [8]. EXFOR [9] holds 67 experiments in this range. This work uses all of them to produce a purely data-driven `MF4` and `MF34`, and scores the result and the libraries against the database under an identical metric.

Table 1. Angular-distribution covariance (`MF34`) available for ^{56}Fe elastic scattering in the 847 keV–4 MeV range.

Evaluation	Orders	Energy structure
JEFF-4.0*	$\ell = 1$–6	13 groups†, uncorrelated
JENDL-5	$\ell = 1$	4 groups, correlated
ENDF/B-VIII.1	none	—
This work	$\ell = 1$–6	526 groups, correlated

* From JEFF-3.3 (2005). † 10 above 1.4 MeV for $\ell = 5, 6$.

2 Database and comparison metric

The database consists of 67 EXFOR experiments (subentries) measuring ^{56}Fe or natural-iron elastic differential cross sections between 847 keV and 4 MeV, 46 742 angular points in total. Most campaigns used natural iron, which is more than 91% ^{56}Fe, and the enriched datasets show no systematic offset with respect to them. The pool is very unevenly distributed: Cierjacks 1978 [10] alone supplies 28 631 points, about 61% of the total.

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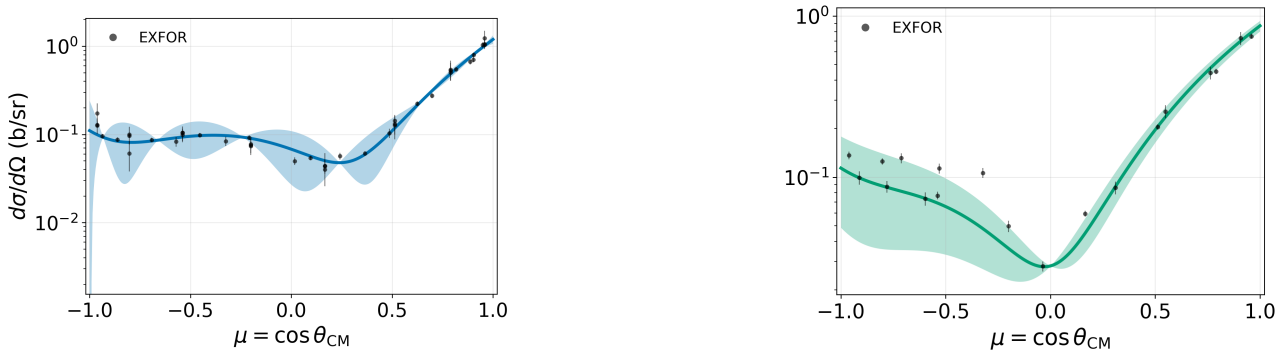


Figure 1. Contribution of a single Legendre order to the propagated $\pm 1\sigma$ band on the ^{56}Fe elastic differential cross section. Left: $\ell = 6$ in JEFF-4.0 at $E_n = 2.800$ MeV. Right: $\ell = 1$ in JENDL-5 at $E_n = 1.571$ MeV. Black markers are EXFOR points in the corresponding energy bin.

The uncertainty of each measured differential cross-section value (one angle, one energy) is split into a statistical component, uncorrelated between values, and a systematic one shared by all values of an experiment. The split is read once from the `ERR-ANALYS` description of each EXFOR entry and recorded in a per-experiment manifest. Where no split is reported, the systematic takes a 5% default, the median of the documented values. It is taken by 31 of the 67 experiments: thirty that report no decomposition, holding only 5% of the points, and Cierjacks, which reports one 7–12% total lumping flux, efficiency, sample and background corrections. The data covariance of experiment g is then rank one plus diagonal,

$$\Sigma^{(g)} = D^{(g)} + u^{(g)} u^{(g)\mathsf{T}}, \tag{1}$$

with $D^{(g)}_{ii} = \sigma^2_{\mathrm{stat},i}$ and $u^{(g)}_i = \sigma_{\mathrm{sys},i}$. Different experiments are taken as uncorrelated.

As the elastic cross section fluctuates here faster than the experimental resolution, its tabulated value (`MF3`) at the nominal energy misses the measured magnitude and would mask the angular shape, so the comparison uses the shape-normalized form

$$\frac{d\sigma}{d\Omega}(\mu) = c_0(E)\Big[1 + \sum_{\ell=1}^{L} (2\ell+1)\, a_\ell(E)\, P_\ell(\mu)\Big], \tag{2}$$

where the a_ℓ are the `MF4` Legendre coefficients and $c_0(E)$ replaces the magnitude: it is fitted at each energy to the measurements that drove each evaluation, Kinney and Smith for the libraries and all experiments in that energy bin for this work. The goodness of fit of experiment g is measured by the Mahalanobis statistic

$$\chi^2_g = \mathbf{r}_g^{\mathsf{T}}\, C_g^{-1}\, \mathbf{r}_g, \qquad C_g = \Sigma^{(g)} + \Sigma^{(g)}_{\mathrm{eval}}, \tag{3}$$

where $\mathbf{r}_g$ is the residual against the evaluation and $\Sigma^{(g)}_{\mathrm{eval}}$ is the library's `MF34` propagated onto the measured (E, μ) points. Scaling $u^{(g)}$ by the prediction avoids Peelle's Pertinent Puzzle [11]. The full covariance is essential here: a 5% calibration offset shared by a hundred points of one experiment is a single 1σ event, which a diagonal χ^2 would count a hundred times. The reported χ^2/N sums over experiments, N being the number of points. Because $c_0(E)$ comes from the data, the evaluated cross section does not enter, and two terms are left out on purpose so that the metric isolates the angular distribution and its covariance: the magnitude uncertainty (`MF33`), which differs between libraries and would hide the effect of their different `MF34`, and the folding with the energy resolution of each experiment, which needs time-of-flight parameters that EXFOR often lacks and can match only one experimental configuration at a time. These choices, and the offsets between campaigns beyond their declared uncertainties, set the absolute level of χ^2/N, so $\chi^2/N = 1$ is not a usable reference; the meaningful result is the comparison between evaluations.

3 Evaluation pipeline

Energy grid.

Bins are built from the union of the energies reported by Kinney below 2.5 MeV and by Pirovano [12] above it (83 energies, against 29 for Smith), giving 1738 bins. When an experiment reports several energies within a bin, only the angular distribution closest to the bin center enters the fit, so that a campaign with a dense energy mesh (mainly Cierjacks) does not dominate the bin. This leaves 23 253 of the 46 742 points for the fit; the χ^2 scores all of them.

Nominal fit.

In each bin the coefficients are determined by ridge-regularized generalized least squares,

$$\hat{\mathbf{c}} = (X^{\mathsf{T}}\tilde{\Sigma}^{-1}X + \Lambda)^{-1} X^{\mathsf{T}}\tilde{\Sigma}^{-1}\mathbf{y}, \tag{4}$$

with $X_{i\ell} = P_\ell(\mu_i)$, $\Lambda = \mathrm{diag}(\lambda \ell^p)$, $\lambda = 10^{-4}$ and $p = 4$, which penalizes the high orders. The covariance $\tilde{\Sigma}$ is that of Eq. (1), rescaled by a Kish effective-sample-size factor per experiment and energy, so that the many angular points of one experiment at one energy, correlated through their shared systematic, count roughly as one observation; no experiment may exceed 80% of the weight of a bin. The order of the fit is selected per bin by the Akaike information criterion, AIC $= \chi^2 + 2k$ with k the trace of the hat matrix, up to $L_{\max} = 6$. Where campaigns disagree

and the points scatter about the fit beyond their declared uncertainties, the statistical uncertainties are scaled up by factors s_b within three angular bands (forward, mid, backward), estimated from that scatter by its median absolute deviation and iterated with the fit.

Variance pass.

The first Monte Carlo pass estimates the marginal variance of every coefficient in every bin. In each sample, every experiment receives a log-normal normalization factor drawn independently per bin (only the variance of each bin is sought here), each point receives independent Gaussian noise of width $s_b\,\sigma_{\mathrm{stat}}$, the scaled uncertainty of the nominal fit, and the bin is refitted with the Kish weights and c_0 held at their nominal values. The fitted order is redrawn in each sample from the Akaike weights [13], which carries the model-form uncertainty into the ensemble. Ten thousand samples are drawn, with the differential cross section kept non-negative at all angles.

Correlation pass.

The second pass produces the full correlation matrix R_{KW} (kernel weight), across orders and across energy, with the order fixed at the AIC choice. It differs from the first in one decisive way: the normalization factor drawn for an experiment is applied to every bin the experiment reaches, weighted by the overlap between the bin and its Gaussian time-of-flight resolution kernel. Each measured value also enters every bin its resolution kernel reaches with the same noise draw, so adjacent bins sharing measurements move together. No correlation length is imposed.

Assembly.

Variances from the first pass and correlations from the second are combined by the congruence transform

$$C = \mathrm{diag}(\sigma)\; R_{\mathrm{KW}}\; \mathrm{diag}(\sigma). \tag{5}$$

Taking the whole of R_{KW} from a single pass keeps C positive semi-definite by construction. The covariance on the fine energy grid, 1738 bins across six orders, is collapsed onto an adaptive multigroup grid by merging adjacent bins whose a_1 correlation reaches 0.85 while the ratio of their a_1 standard deviations stays below 5. Averaging lowers the variance below that of the merged bins, so each group is scaled up to a percentile (67th–85th) of its fine-bin variances, giving 526 groups shared by all six orders.

4 Results

Table 2. χ^2/N under the full budget $\sigma_{\mathrm{exp}} \oplus \sigma_{\mathrm{eval}}$; bold is lowest.

Subset	N	JEFF-4.0	JENDL-5	This work
All	46 742	9.6	15.1	**7.5**
Kinney only	13 208	3.0	**0.53**	8.9
No Kinney	33 534	12.2	20.9	**7.0**
Cierjacks only	28 631	11.4	15.7	**7.4**
No Kinney, no Cierjacks	4 903	17.1	51.1	**4.7**

Central values.

The fitted coefficients agree with the libraries generally within 1σ, yet the differences show in the metric. With the EXFOR uncertainties alone, χ^2/N on the whole database is 14.5 for this work against 21.8 and 22.4 for the libraries, and this work is the lowest on every subset of Table 2 except Kinney alone.

Covariance.

The a_1 uncertainty of this work (Fig. 2a) generally falls between those of JEFF-3.3 and JENDL-5; it is small where a single campaign, often Kinney, constrains the fit, and peaks where several campaigns meet and disagree. Its correlation (Fig. 2b) shows two positive blocks, below 2.5 MeV and above 2 MeV, each set by the shared normalization of the campaigns that dominate that range (Kinney and Cierjacks, then Pirovano and Smith). The few strong negative correlations between adjacent bins, around 1.0–1.8 MeV, appear where neighboring bins prefer different Legendre orders.

Goodness of fit.

With each evaluation's `MF34` included (Table 2), this work is the lowest on the whole database and, scoring one experiment at a time, on 60 of the 67 experiments, against 4 for JEFF-4.0 and 3 for JENDL-5. Its margin over JEFF-4.0 is modest, and the subsets show why.

Both libraries were fitted to Kinney, and on that campaign alone the ordering reverses. JENDL-5 fits the points as reported and falls below the uncertainties Kinney declares for itself; JEFF-4.0 stays above unity because the CIELO evaluation corrected the backward-angle normalization of those data. This work weights Kinney no more than any other experiment. Without Kinney, the libraries worsen while this work improves. Cierjacks still supplies 85% of the remaining points and follows the same pattern. Once it is removed too, the 65 remaining independent experiments separate the evaluations most clearly. The agreement of the libraries with the full database therefore rests on the campaign they were fitted to and on a relatively favorable match to Cierjacks, not on uniform agreement across the data.

5 Conclusions

This work delivers a reproducible, purely data-driven and mutually consistent `MF4` and `MF34` for ^{56}Fe elastic scattering between 847 keV and 4 MeV from the whole EXFOR database, with the full method documented in [14]. The uncertainty is spread across the fitted Legendre orders, with an inter-energy correlation set by the resolution and normalization of the experiments themselves, on a 526-group structure. Under an identical metric, it reaches the lowest χ^2/N on 60 of the 67 experiments and on every subset once Kinney 1976 is set aside.

Three limitations remain. (i) The χ^2/N stays well above unity for all three evaluations; its absolute level depends on the metric, but the uncertainties reported here

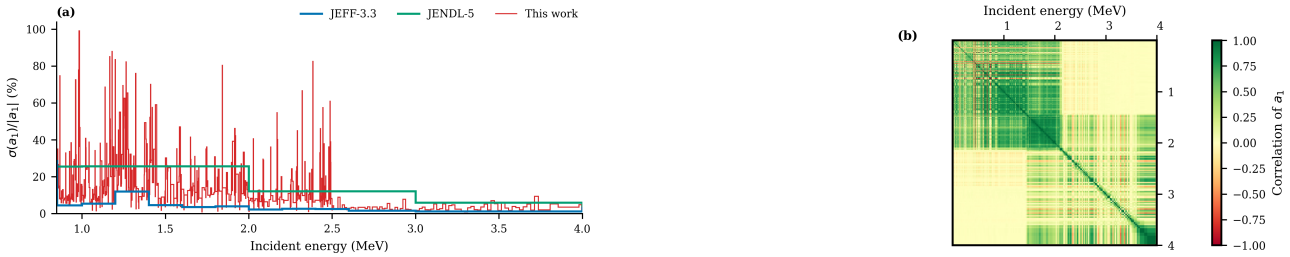


Figure 2. (a) Relative uncertainty of a_1 in JEFF-3.3, JENDL-5 and this work. (b) Energy correlation of a_1 in this work.

may still understate the spread the measurements imply. (ii) Most bins rest on two or three independent campaigns that often disagree by more than their stated uncertainties, so the between-experiment spread that the covariance is meant to capture is itself poorly constrained. For a better covariance estimate, the most valuable new measurement would therefore match Kinney's density in energy and angle. (iii) The campaigns differ in energy resolution, and combining them bin by bin costs either the structure resolved by the finest or the averaging carried by the coarsest. An estimator that carries each experiment's resolution inside the model would use both without degrading either.

Their propagation to shielding benchmarks and vessel fluence is reported in [15].

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